

Curriculum vitae

Hugo Pomares-Millan, MD, MPH, PhD

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/ [GitHub: hpomares](#)

Research overview: My academic and professional background encompasses medicine, genetic epidemiology, and public health, with expertise spanning clinical practice, medical affairs, and regulatory science. My research integrates large-scale epidemiological studies and machine learning methodologies to investigate the impact of lifestyle and environmental exposures on genetically susceptible populations. A key objective of my work is to translate these findings into clinically applicable tools to improve risk stratification, facilitate early detection, and advance precision prevention strategies. Through active collaborations with international consortia, I aim to generate high-quality evidence that informs personalized interventions and shapes public health policies.

I. EDUCATION

<u>DATE</u>	<u>INSTITUTION</u>	<u>DEGREE</u>
2018 – 2022	Lund University, Lund, Sweden	Doctoral Studies (PhD); Received: October 2022 Supervisor(s): Drs. Paul Franks/ Giuseppe N. Giordano / Anders Rosengren Thesis: “Optimizing Exposome-wide Assessments in Cardiometabolic Risk”
2016 – 2018	Lund University, Lund, Sweden	Master’s in Public Health (MPH); Received: Summer 2018
2011 – 2013	National Autonomous University of Mexico (UNAM), Mexico City	Specialization in Occupational Health (Honors)

2005 – 2011	National Autonomous University of Mexico (UNAM), Mexico City	Medical Doctor (MD)
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II. POSTDOCTORAL TRAINING

<u>DATE</u>	<u>INSTITUTION</u>	<u>ROLE</u>
2023 – 2025	Dartmouth College, Geisel School of Medicine, Hanover, NH	Postdoctoral Research Associate Supervisor(s): Drs. Michael N. Passarelli / Siming Zhao / Prof. Margaret R. Karagas
2022	Dartmouth College, Geisel School of Medicine, Hanover, NH	Visiting Research Scientist Supervisor(s): Prof. Brock Christensen

III. BIBLIOGRAPHY

<u>METRIC</u>	<u>VALUE</u>
Total Number of Peer Reviewed Journal Articles:	27
Total Number of 1 st author:	5
Total Number of Last/Senior Author:	1
Google Scholar H-Index:	10
Google Scholar Number of Citations:	530

IV. PEER-REVIEWED PUBLICATIONS

IV.A First/Senior authorship (*contributed equally)

Pomares-Millan H, Saxby MS, Al-Mashadi SD, Karagas MR, Passarelli MN.

Dietary glycemic index, glycemic load, sugar, and fiber intake in association with breast cancer risk: An updated meta-analysis (In press). *Nutrition Reviews*. 2025.

Prinz N*, **Pomares-Millan H***, Dannemann A, Giordano GN, Joisten C, et al.

Who profits most from outpatient lifestyle intervention? – An IMI-SOPHIA study on the heterogeneous treatment response of children and adolescents living with overweight/obesity. *Obesity*. 2023 Aug 6. DOI: 10.1002/oby.23844

Pomares-Millan H, Poveda A, Atabaki-Pasdar N, Johansson I, Björk J, Ohlsson M, Giordano GN, Franks PW.

Predicting Sensitivity to Adverse Lifestyle Risk Factors for Cardiometabolic Morbidity and Mortality. *Nutrients*. 2022 Jan;14(15):3171. DOI: 10.3390/nu14153171

Pomares-Millan H, Atabaki-Pasdar N, Coral D, Johansson I, Giordano GN, Franks PW.

Estimating the Direct Effect between Dietary Macronutrients and Cardiometabolic Disease, Accounting for Mediation by Adiposity and Physical Activity. *Nutrients*. 2022 Mar 13;14(6):1218. DOI: 10.3390/nu14061218

Franks PW, **Pomares-Millan H**.

Next-generation epidemiology: the role of high-resolution molecular phenotyping in diabetes research. *Diabetologia*. 2020 Dec;63(12):2521-32. DOI: 10.1007/s00125-020-05246-w

Mutie PM*, **Pomares-Millan H***, Atabaki-Pasdar N, Jordan N, Adams R, Daly NL, Tajés JF, Giordano GN, Franks PW.

An investigation of causal relationships between prediabetes and vascular complications. *Nature communications*. 2020 Sep 14;11(1):1-1. DOI: 10.1038/s41467-020-18386-9

IV.B Co- authorship

Passarelli MN, McDonald JG, Thompson BM, **Pomares-Millan H**, et al.

Plasma concentrations of multiple oxysterols and risk of colorectal adenomas. *Cancer Prevention Research*. 2024;17(11), pp.517-524. DOI: 10.1158/1940-6207.CAPR-24-0108

Dhanasekaran B, Morton W R, Santhakumar V, Nakabuye M, **Pomares-Millan H**, et al.

Impact of individual and environmental factors on dietary or lifestyle interventions to prevent type 2 diabetes development: a systematic review. *Communications Medicine*. 2023;3(133). DOI: 10.1038/s43856-023-00363-0

Tobias DK, Merino J, Ahmad A, Aiken C, ...**Pomares-Millan H**, et al.

Second international consensus report on gaps and opportunities for the clinical translation of precision diabetes medicine. *Nature Medicine*. 2023 Oct;29(10):2438-2457. DOI: 10.1038/s41591-023-02502-5.

Patel SR, Riano I, Abuali I, Ai A, Geiger G, ...**Pomares-Millan H**, et al.

Race/Ethnicity and Gender Representation in Hematology and Oncology Editorial Boards: What is the State of Diversity? *The Oncologist*. 2023 Apr 29:oyad103. DOI: 10.1093/oncolo/oyad103

Mutie PM, **Pomares-Millan H**, et al.

Investigating the causal relationships between excess adiposity and cardiometabolic health in men and women. *Diabetologia*. 2023 Feb;66(2):321-35. DOI: 10.1007/s00125-022-05811-5

Coral DE, Fernandez-Tajes J, Tsereteli N, **Pomares-Millan H**, et al.

A phenome-wide comparative analysis of genetic discordance between obesity and type 2 diabetes. *Nature Metabolism*. 2023 Feb;5(2):237-47. DOI: 10.1038/s42255-022-00731-5

Riano I, **Pomares-Millan H**, et al.

Evaluating Internal Medicine Residents' Awareness on Cancer Survivorship Care Plan: A Pilot Survey. *Journal of Community Hospital Internal Medicine Perspectives*. 2022;12(6):12. DOI: 10.55729/2000-9666.1126

Atabaki-Pasdar N, Ohlsson M, Viñuela A, Frau F, **Pomares-Millan H**, et al.

Predicting and elucidating the etiology of fatty liver disease: A machine learning modeling and validation study in the IMI DIRECT cohorts. *PLoS medicine*. 2020 Jun 19;17(6):e1003149. DOI: 10.1371/journal.pmed.1003149

IV.C Consortia-level authorship

(<https://www.ncbi.nlm.nih.gov/myncbi/12WEerr8joKAU/bibliography/public/>)

Naylor RN, Patel KA, Kettunen JL, Männistö JM, Støy J, Beltrand J, ...**ADA/EASD PMDI**, et al.

Precision treatment of beta-cell monogenic diabetes: a systematic review. *Communications medicine*. 2024 Jul 18;4(1):145. DOI: 10.1038/s43856-024-00556-1

Takele WW, Vesco KK, Josefson J, Redman LM, Hannah W, ...**ADA/EASD PMDI**, et al.

Effective interventions in preventing gestational diabetes mellitus: A systematic review and meta-analysis. *Communications medicine*. 2024 Apr 20;4(1):75. DOI: 10.1038/s43856-024-00491-1

Felton JL, Redondo MJ, Oram RA, Speake C, Long SA, ...**ADA/EASD PMDI**, et al.

Islet autoantibodies as precision diagnostic tools to characterize heterogeneity in type 1 diabetes: a systematic review. *Communications medicine*. 2024 Apr 6;4(1):66. DOI: 10.1038/s43856-024-00478-y.

Ahmad A, Lim LL, Morieri ML, Tam CH, Cheng F, ...**ADA/EASD PMDI**, et al.

Precision prognostics for cardiovascular disease in Type 2 diabetes: a systematic review and meta-analysis. *Communications medicine*. 2024 Jan 22;4(1):11. DOI: 10.1038/s43856-023-00429-z.

Semnani-Azad Z, Gaillard R, Hughes AE, Boyle KE, Tobias DK, ...**ADA/EASD PMDI**, et al.

Precision stratification of prognostic risk factors associated with outcomes in gestational diabetes mellitus: a systematic review. *Communications medicine*. 2024 Jan 12;4(1):9. DOI: 10.1038/s43856-023-00427-1.

Francis EC, Powe CE, Lowe WL Jr, White SL, Scholtens DM, ...**ADA/EASD PMDI**, et al.

Refining the diagnosis of gestational diabetes mellitus: a systematic review and meta-analysis. *Communications medicine*. 2023 Dec 18;3(1):185. DOI: 10.1038/s43856-023-00393-8.

Misra S, Wagner R, Ozkan B, Schön M, Sevilla-Gonzalez M, ...**ADA/EASD PMDI**, et al.

Precision subclassification of type 2 diabetes: a systematic review. *Communications medicine*. 2023 Oct 5;3(1):138. DOI: 10.1038/s43856-023-00360-3.

- Benham JL, Gingras V, McLennan NM, Most J, Yamamoto JM, ...**ADA/EASD PMDI**, et al.
Precision gestational diabetes treatment: a systematic review and meta-analyses. *Communications medicine*. 2023 Oct 5;3(1):135. DOI: 10.1038/s43856-023-00371-0.
- Felton JL, Griffin KJ, Oram RA, Speake C, Long SA, ...**ADA/EASD PMDI**, et al.
Disease-modifying therapies and features linked to treatment response in type 1 diabetes prevention: a systematic review. *Communications medicine*. 2023 Oct 5;3(1):130. DOI: 10.1038/s43856-023-00357-y.
- Young KG, McInnes EH, Massey RJ, Kahkoska AR, Pilla SJ, ...**ADA/EASD PMDI**, et al.
Treatment effect heterogeneity following type 2 diabetes treatment with GLP1-receptor agonists and SGLT2-inhibitors: a systematic review. *Communications medicine*. 2023 Oct 5;3(1):131. DOI: 10.1038/s43856-023-00359-w.
- Murphy R, Colclough K, Pollin TI, Ikle JM, Svalastoga P, ...**ADA/EASD PMDI**, et al.
The use of precision diagnostics for monogenic diabetes: a systematic review and expert opinion. *Communications medicine*. 2023 Oct 5;3(1):136. DOI: 10.1038/s43856-023-00369-8.
- Lim S, Takele WW, Vesco KK, Redman LM, Hannah W, ...**ADA/EASD PMDI**, et al.
Participant characteristics in the prevention of gestational diabetes as evidence for precision medicine: a systematic review and meta-analysis. *Communications medicine*. 2023 Oct 5;3(1):137. DOI: 10.1038/s43856-023-00366-x.
- Jacobsen LM, Sherr JL, Considine E, Chen A, Peeling SM, ...**ADA/EASD PMDI**, et al.
Utility and precision evidence of technology in the treatment of type 1 diabetes: a systematic review. *Communications medicine*. 2023 Oct 5;3(1):132. DOI: 10.1038/s43856-023-00358-x.
- Semple RK, Patel KA, Auh S, Brown RJ...**ADA/EASD PMDI**, et al.
Genotype-stratified treatment for monogenic insulin resistance: a systematic review. *Communications medicine*. 2023 Oct 5;3(1):134. DOI: 10.1038/s43856-023-00368-9.
- Brown AA, Fernandez-Tajes JJ, Hong MG, Brorsson CA, Koivula RW, ... **IMI DIRECT Consortium** et al.
Genetic analysis of blood molecular phenotypes reveals common properties in the regulatory networks affecting complex traits. *Nature Communications*. 2023 Aug 21;14(1):5062. DOI: 10.1038/s41467-023-40569-3.
- Allesøe RL, Lundgaard AT, Hernández Medina R, Aguayo-Orozco A, Johansen J, ... **IMI DIRECT Consortium** et al.
Discovery of drug-omics associations in type 2 diabetes with generative deep-learning models. *Nature Biotechnology*. 2023 Mar;41(3):399-408. DOI: 10.1038/s41587-022-01520-x.
- Tura A, Grespan E, Göbl CS, Koivula RW, Franks PW, ...**IMI DIRECT Consortium** et al.

et al.

Profiles of glucose metabolism in different prediabetes phenotypes, classified by fasting glycemia, 2-hour OGTT, glycated hemoglobin, and 1-hour OGTT: an IMI DIRECT study. *Diabetes*. 2021 Sep 1;70(9):2092-106. DOI: 10.2337/db21-0227

Bizzotto R, Jennison C, Jones AG, Kurbasic A, Tura A, ...**IMI DIRECT Consortium** et al.

Processes underlying glycemic deterioration in type 2 diabetes: an IMI DIRECT study. *Diabetes Care*. 2021 Feb 1;44(2):511-8 DOI: 10.2337/dc20-1567

Wilman HR, Parisinos CA, Atabaki-Pasdar N, Kelly M, ...**IMI DIRECT Consortium** et al.

Genetic studies of abdominal MRI data identify genes regulating hepcidin as major determinants of liver iron concentration. *Journal of Hepatology*. 2019 Sep;71(3):594-602. DOI: 10.1016/j.jhep.2019.05.032.

Dawed AY, Zhou K, van Leeuwen N, Mahajan A, Robertson N, ...**IMI DIRECT Consortium** et al.

Variation in the plasma membrane monoamine transporter (PMAT)(encoded by SLC29A4) and organic cation transporter 1 (OCT1)(encoded by SLC22A1) and gastrointestinal intolerance to metformin in type 2 diabetes: an IMI DIRECT study. *Diabetes Care*. 2019 Jun 1;42(6):1027-33. DOI: 10.2337/dc18-2182

IV.D Work in progress

Pomares-Millan H, Stella K, Baris D, Schwenn M, Johnson A, Rothman N, Silverman DT, Karagas MR, Passarelli MN.

Lifetime Water Arsenic, Genetic Susceptibility, And Bladder Cancer In The New England Bladder Cancer Study (*under review*) *JNCI Cancer Spectrum* 2025.

Pomares-Millan H, Zhao S, Karagas MR.

Interaction between arsenic in water, urinary arsenic species, and bladder cancer risk in the New Hampshire Bladder Cancer study (*under preparation*) 2025.

Julián-Serrano S, **Pomares-Millan H**, Noel ES, Mangano KM, Li W, Fouhy L, Xiyuan Zhang X, Ordovás JM, Lai C-Q, Tucker KL.

Sweet taste sensitivity genetic risk score and alignment to diet quality: Findings from the Boston Puerto Rican Health Study (*under preparation*) 2025.

Pomares-Millan H, Karagas MR.

Lipid-lowering therapies, adiposity traits, and breast cancer risk: A mediation analysis using a multivariable mendelian randomization framework (*under preparation*) 2024.

V. LICENSES & CERTIFICATIONS

<u>YEAR</u> <u>ISSUED</u>	<u>INSTITUTION</u>	<u>LICENSE</u>
2016	National Board of Health and Welfare (Socialstyrelsen), Sweden	Doctor of Medicine
2013	Secretariat of Public Education, Mexico	Occupational Health specialist (8404238)
2011	Secretariat of Public Education, Mexico	Doctor of Medicine (6988198)

VI. WORK EXPERIENCE

<u>DATE</u>	<u>INSTITUTION</u>	<u>POSITION</u>
2018	Genetic and Molecular Epidemiology unit, Lund University, Lund, Sweden	<i>Research assistant:</i> Between the Master's and Doctoral programs, I conducted literature reviews, collected and analyze data descriptively. Prepare scientific materials and presentations for grant agencies and foundations.
2015 – 2016		<i>Manager of the New Molecules department:</i> Responsible for technical, scientific, and medical/regulatory affairs of new drug applications (NDA), investigational new drugs (IND), and meetings with the industry. Prepare technical guidance in accordance with international guidelines for preclinical and clinical dossier.
2014 – 2015	COFEPRIS Regulatory agency, Ministry of Health, Mexico	<i>Medical Officer:</i> Conducting medical evaluations to authorization dossier, technical and administrative variations and extensions of allopathic drugs, biotechnological products, and vaccines; Medical judge for external legal cases; Implemented procedures for COFEPRIS certification with Pan-American Health Organization (PAHO) and World Health Organization (WHO) 2014
2012 – 2014		<i>Medical Reviewer:</i> Conducting regulatory drug assessments, evaluation of the medical dossier of

allopathic drugs, biotechnological products, and vaccines.

2010 – 2011 Unilever Mexico S de R.L.

Occupational Health trainee: Assist in health risk assessments; Raise awareness of good practices to prevent work-related injuries, musculoskeletal disorders, and stress by on-site campaigns.

VII. TEACHING ACTIVITIES

<u>TERM</u>	<u>INSTITUTION</u>	<u>COURSE/ACTIVITY</u>	<u>ROLE</u>	<u>COURSE SIZE</u>
Fall 2024	Dartmouth College	MDED 118 Patients & Populations	Instructor	13
Fall 2024; Fall 2023	Dartmouth College	Foundations of Epidemiology II (QBS 131/Biol 073): Case-control study design	Invited Lecturer	11
Spring 2020; Spring 2019	Lund University	Master's in Public Health: Epidemiology study designs	Invited Lecturer	23
2015	Ministry of Health, Mexico	Course: Workshop of medical review and writing, effective literature research, and critical analysis of research papers	Instructor	25
2014	Ministry of Health, Mexico	Course: Industry Guidance – OTC Switch and New Molecules	Instructor	30

VIII. RESEARCH ADVISING

<u>DATE</u>	<u>INSTITUTION</u>	<u>ADVISEE/MENTEE</u>	<u>PROGRAM</u>
Winter 2023	University of Massachusetts, Lowell	Sachelly Julián-Serrano	Research Advising, Epidemiology ScD Candidate

IX. MEMBERSHIP & LEADERSHIP

<u>DATE</u>	<u>INSTITUTION</u>	<u>ROLE</u>
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2023 – 2024	Society for Epidemiologic Research (SER)	Team Mentoring program coordinator
2020	ADA/EASD initiative: Precision prevention working group (PMDI)	Member
2018 – 2022	IMI-DIRECT (Diabetes research on patient stratification) consortium	Member
2018 – 2022	IMI-SOPHIA (Stratification of obese phenotypes to optimize future obesity therapy) consortium	Member
2015	VII Latin Association of Nonprescription Medications (ILAR), Workshop about OTC switch, Brazil	Representative
2015	VI Latin Association of Nonprescription Medications (ILAR), Workshop about OTC switch, Argentina	Representative
2014	Seminar “GCP and ethical aspects of research, development of a new thematic network in the vicinity of the network of competent authorities on drugs in Latin America”, Colombia	Representative
2014	Vaccines WHO Workshop on the regulatory requirements for the preclinical / non-clinical and clinical evaluation of vaccines (human, live attenuated and inactivated), Thailand	Representative

X. POSTER PRESENTATIONS

<u>YEAR</u>	<u>TITLE</u>	<u>MEETING</u>
2024	<u>Pomares-Millan H.</u> Koutros S, Rothman N, Baris D, Schwenn M, Johnson A, Silverman DT, Karagas MR; Passarelli MN. Gene-environment interactions for lifetime water arsenic exposure and bladder cancer risk in the New England Bladder Cancer study. <i>Cancer Research</i> 84, no. suppl 6:6140 (2024); DOI: 10.1158/1538-7445.AM2024-6140	American Association for Cancer Research (AACR), San Diego, CA
2013	<u>Pomares-Millan H.</u> Working Conditions of the University Professor Associated to Burnout Syndrome and Mood Disorder	American Occupational Health

Conference (AOHC),
Orlando, FL

XI. CONFERENCE PRESENTATIONS

<u>DATE</u>	<u>TALK</u>	<u>ROLE</u>
2024	11 th Research Forum of the Network of Postgraduate Studies in Occupational Health, Mexico	Speaker
2024	4 th Annual Symposium for the Vermont Center for Cardiovascular and Brain Health; University of Vermont Davis Center, USA	Speaker
2021	57 th EASD Annual Meeting of the European Association for the Study of Diabetes, DOI: 10.1007/s00125-021-05519-y	Speaker
2020	56 th EASD Annual Meeting of the European Association for the Study of Diabetes, DOI: 10.1007/s00125-020-05221-5	Speaker
2014	1 st Meeting of National College of Pharmaceutical Chemical Biologists, Mexico	Speaker
2013	1 st Research Forum of the Network of Postgraduate Studies in Occupational Health, Mexico	Speaker

XII. SCHOLARSHIPS & AWARDS

<u>DATE</u>	<u>AWARD</u>	<u>SPONSOR</u>	<u>OBJECTIVE</u>
2024	Postdoc Professional Development Award	Guarini School of Graduate and Advanced Studies, Dartmouth College	Cover expenses AACR 2024 conference
2023	Travel award	Epidemiology department, Geisel School of Medicine, Dartmouth College	Attend Advanced Gene mapping course at Rockefeller university
2018	Travel award	Faculty of Medicine, Lund University	Attend IMI-DIRECT meeting
2017	Travel award	Faculty of Medicine, Lund University	Attend IMI-DIRECT meeting
2016	Travel award	Faculty of Medicine, Lund University	Attend IMI-DIRECT meeting
2016 – 2018	Lund Global Scholarship	Faculty of Medicine, Lund University	Tuition and fees of Master's program

XIII. JOURNAL PEER-REVIEW ACTIVITY

<u>DATES</u>	<u>JOURNAL</u>	<u># Reviews</u>
2022	<i>Nutrition and cancer</i>	4
2023	<i>Oncology Reviews</i>	1
2022	<i>Nutrients</i>	3

XIV. MEMBERSHIPS PROFESSIONAL SOCIETIES

<u>YEAR</u>	<u>INSTITUTION</u>
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2022	Member of American Association for Cancer Research (AACR) ID: 129074
2022	Member of the Society for Epidemiologic research (SER)
2014	Founding Member of the Colegio Ramazzini de Mexico, A.C.

XV. STATISTICAL & PROGRAMMING SKILLS

<u>SOFTWARE/PACKAGE</u>	<u>TASKS</u>
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R	Data processing, wrangling, visualization, and exploratory analysis. Machine Learning models: classification, regression, clustering, feature engineering. Observational inferential methods; Linear, logistic regression, mixed effects, mediation analysis, and repeated measures methods (i.e., Hierarchical models)
Microsoft	Word, PowerPoint, Excel, Teams
UNIX-like command tools	Genetic analysis (GCTA, FastLMM, BOLT-LMM, PLINK, MR, LDSC)

XVI. OTHER SKILLS

Medical writing: Writing and editing of research collaboration proposals and data analysis plans (IMI-DIRECT consortia). During my position at the Ministry of Health, I supervised writing style and tone of material for clinicians (prescribing information, PDR) and supervise OTC labels for target audience (lay language for patients).

Project and Team Management: Manage timelines and deliverables for multiple NDA and IND projects. Coordinate a cross-functional team to meet deadlines for new drug approvals. Under my supervision the team had 6 supervisees (2 medical doctors, 3 chemists, 1 administrative support). In 2015, we were the first country to revise the IND dossier and grant approval to Dengue vaccine (Sanofi).

Languages: I speak English (functionally native), Spanish (native), Swedish (intermediate).